

Sang-Woo Son

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Research Interests

LLM agents that hold up in production, including tool-calling control and guardrails, multi-agent orchestration, retrieval-grounded evaluation, and agents that act over real-time voice.

Publications

INTERNATIONAL

Sang-Woo Son, Hyeong-seob Kim, Hyeonsang Kim, Hyun-woo Cho, Jinmo Kim

Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

Accepted to EMNLP 2026 Industry Track.

[GITHUB](#)

[VIEW DETAILS](#)

Hyeong-seob Kim, Sang-Woo Son, Hyun-woo Cho, Hyeonsang Kim, Jinmo Kim

WIGVO: Real-Time Bidirectional Speech Translation over Legacy PSTN Calls via Dual-Session Echo Gating

In Proceedings of ACL 2026 System Demonstrations.

[PAPER](#)

[VIDEO](#)

[GITHUB](#)

Sang-Woo Son, Jae Hyun Rho, Jae-Won Lee, Hye-Min Lee, Youngwoo Lee, Hyeon-Beom Choi, Eeksu Leem, Jeongwook Seo

Implementation of an IoT Cocktail Machine Using ChatGPT API and ConvAnalyser in the Metaverse

In Proceedings of IEEE MetaCom 2024, pp. 305–309.

[PAPER](#)

Jae-Won Lee, Youngwoo Lee, Hyeon-Beom Choi, Sang-Woo Son, Eeksu Leem, Jeongwook Seo

A Metaverse Avatar Teleport System Using an AIoT Pose Estimation Device

In Proceedings of IEEE MetaCom 2023, pp. 698–703.

[PAPER](#)

Sang-Woo Son, Seung-Mi Ham, Min-Jun Kim, Jae-Hyun Rho, Jae-Won Lee, Min-Gu Kang, Jeongwook Seo

Development of Digital Twins of IoT Motion and Vibration Sensors for Attendance Checking and Announcements in the Metaverse

KICS Summer Conference 2022, pp. 327–328 (in Korean).

[PAPER](#)

Work Experience

Jan 2026 - Present

WIGTN

AI Research Engineer, AI Research & Engineering Company

- Develop WIGVO, a production real-time phone translation service over PSTN, improving translation quality through the echo gate and VAD pipeline, operator speaker identification, and a noise-robust audio front end.

[VIEW DETAILS](#)

- Built call observability and evaluation: per-turn Langfuse tracing, a live pipeline monitor, and an LLM-judge translation-quality evaluation with self-validation.

Mar 2026 - Jun 2026

Soundmind

Team Lead, AX Division

- Led the design and implementation of a real-time voice AI agent engine for insurance sales-compliance monitoring, handling simple intents with rule-based guardrails and routing only complex ones (risky utterances, complaints, answer reversals) to LLM tool calling. [VIEW DETAILS](#)
- Served five concurrent calls with a 9B model on a single RTX 3090 at 154 ms P50 and under 300 ms P99, cutting input tokens by 47%.
- Covered non-compliant sales the LLM misses with keyword fallbacks and step-consistency checks, passing all 55 edge-case QA scenarios.

Jan 2025 - Feb 2026

Soundmind

Manager, AX Division

- Built speech AI and LLM products from problem definition and architecture to deployment, across batch STT, keyword spotting, meeting analysis, document RAG, and dialogue agents.
- Designed a large-scale batch STT pipeline on Temporal, Triton, and Faster Whisper, decoupling STT from result callbacks to raise the success rate from 82% to over 95% at 200 requests per minute on two RTX 3090 GPUs.
[VIEW DETAILS](#)
- Developed a KWT-3 custom Korean wake-word model with dual-threshold detection, reaching zero false alarms over 430K+ windows and 96.81% recognition, and cut its size by 75% with TFLite INT8 quantization for edge deployment. [VIEW DETAILS](#)
- Built a LangGraph document RAG system that separates indexing from chat and handles documents up to 50 MB or 100 pages. [VIEW DETAILS](#)
- Built a meeting-analysis platform with Whisper, pyannote diarization, and overlap-weighted speaker-transcript alignment. [VIEW DETAILS](#)

Jan 2023 - Jul 2024

Part-time

Digital Saessak (Digital New Sprouts)

Assistant Instructor, AI Education

- Assisted AI and data-analysis courses and mentored K-12 students in block coding and deep learning.

Research Experience

Sep 2022 - Dec 2024

AID Lab, Hanshin University

- Core Technologies for Autonomous Twins in the Metaverse (ITRC program, MSIT/IITP, led by Sejong University, Sep 2022 - Dec 2024). Built a multimodal depression classifier from text and speech, and an emotion-based IoT cocktail recommender driven by a chatbot.
- Edge-Based Digital Intelligence Platform for Small Businesses (Small Business Smart Technology R&D, MSS/TIPA, led by Ditonic, Oct 2023 - Dec 2024). Built on-device real-time customer behavior recognition, and customer classification and payment prediction from transaction data.
- On-Device IoT Intelligence Engines for Drones and Robots in 5G Environments (ICT R&D program, MSIT/IITP, led by KETI, Sep 2022 - Aug 2023). Built image preprocessing, a behavior-recognition dataset, and an on-device multi-user behavior recognition model.

Hackathons

BYTEDANCE BUILD WITH TRAE HACKATHON 2026 · GRAND PRIZE

WIGENT: Real-Time Multi-Agent Debate Platform

AI agents debate a business idea and turn the outcome into a landing page. Built the debate interface, real-time streaming, and landing page generation from nine templates in a team of four.

[GITHUB](#)[VIEW DETAILS](#)

OBA WEEKENDTHON S1 2026 · TOP 6

MyunZy: AI Mock Interviewer

Reads a real resume and job posting, then runs a spoken mock interview with four interviewer personas that follow up on weak answers. Tracks weaknesses across rounds and allows answer replays.

[GITHUB](#)

GOOGLE CLOUD RAPID AGENT HACKATHON

Custos: Always-On AI Code Review Bot for GitLab

Four reviewer personas analyze each GitLab merge request and post their reviews in under 60 seconds. An hourly proactive lane files prioritized issues on its own and recalibrates from how maintainers react.

[DEVPOST](#)[VIDEO](#)

H0: HACK THE ZERO STACK · VERCEL × AWS

OpenSlot: Zero-Oversell Ticketing Infrastructure

Ticketing infrastructure where each seat is one ledger row changed only by versioned conditional writes. Verified with zero oversells against a simulated 2,000-buyer stampede.

[DEVPOST](#)[VIDEO](#)

GEMINI LIVE AGENT CHALLENGE · GOOGLE

TimeLens: Multimodal AI Museum Curator

A museum curator agent that listens and watches through the visitor's microphone and camera and answers by voice. Audio and vision are processed together in real time on a dual pipeline.

[DEVPOST](#)[VIDEO](#)

INTERNATIONAL ONEM2M HACKATHON 2022 · ENCOURAGEMENT AWARD

Travel Logging with Biometric Data from IoT Wearables

Collects biometric data from IoT wearable devices during a trip and turns it into a travel log. Built on the oneM2M IoT standard for the international oneM2M hackathon.

Education

Sep 2023 - Feb 2025	Hanshin University Convergence of IT Image Data Analytics GPA 4.5 / 4.5 Thesis: Depression Classification in the Metaverse Using Text and Speech Data
Mar 2018 - Aug 2023	Hanshin University IT Transmedia Contents GPA 3.77 / 4.5

Honors & Awards

May 2026	Top 6, OBA Weekendthon S1 Team WIGTN, MyunZy
Mar 2026	Grand Prize, ByteDance Build with TRAE Hackathon Team WIGTN, WIGENT
Nov 2022	Encouragement Award, International oneM2M Hackathon IoT wearable travel logging system
Nov 2022	Honorable Mention, Hanshin Information Science & Convergence Paper Competition CNN-based squat posture analysis system
Oct 2022	Excellence Award, 3rd Hanshin ABC Camp Hackathon Squat posture correction system for home training
Mar 2022	Excellence Award, 2nd Hanshin ABC Camp Hackathon Unity-based promotional content for Osan City

Patents & Software

Dec 2023	IoT-Based Metaverse Management Platform Patent application (Korea) · No. 10-2023-0189803
Nov 2023	Metaverse Emotion Mapping System Using an AIoT Facial-Expression Device Software registration (Korea) · No. C-2023-054487
Nov 2023	LSTM-Based Walking Motion Recognition in Industrial Sites Using YOLOv7 Pose Estimation Software registration (Korea) · No. C-2023-049743